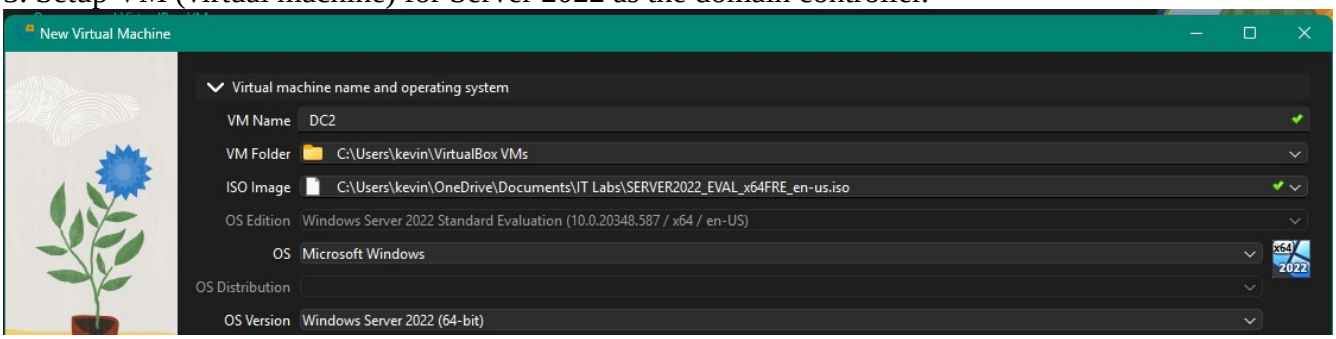
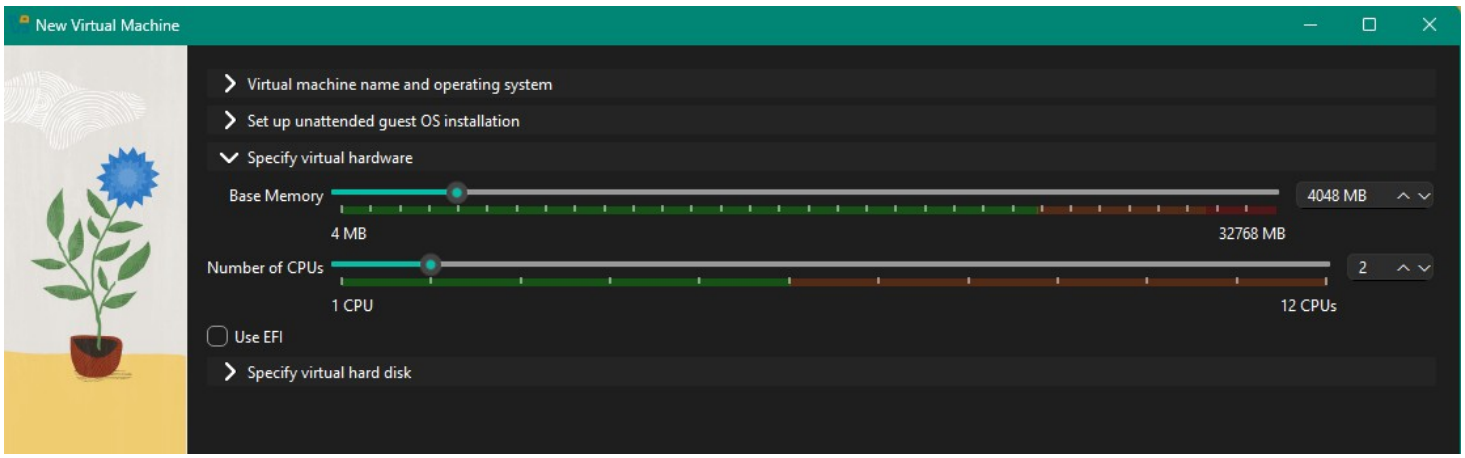
Home Lab Setup

Setup standalone devices

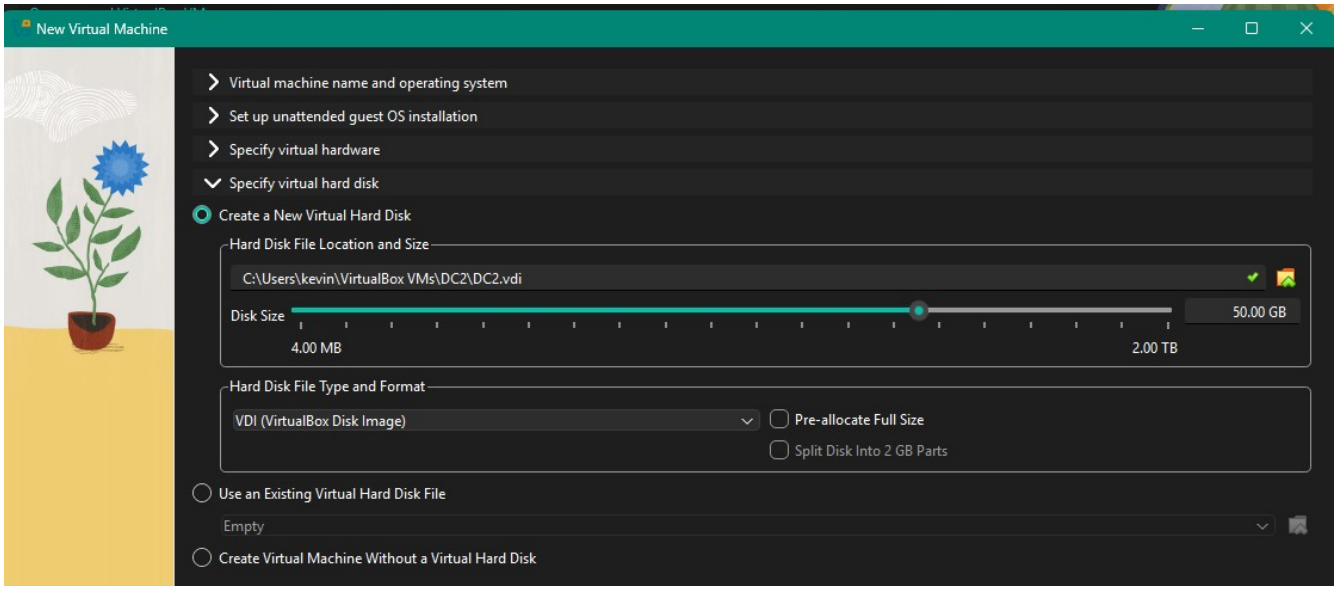
1. Install Virtual Box.
2. Download ISO's for Windows 11 Enterprise and Server 2022 from Microsoft.
3. Setup VM (virtual machine) for Server 2022 as the domain controller.



4GB RAM with 2 CPU cores.

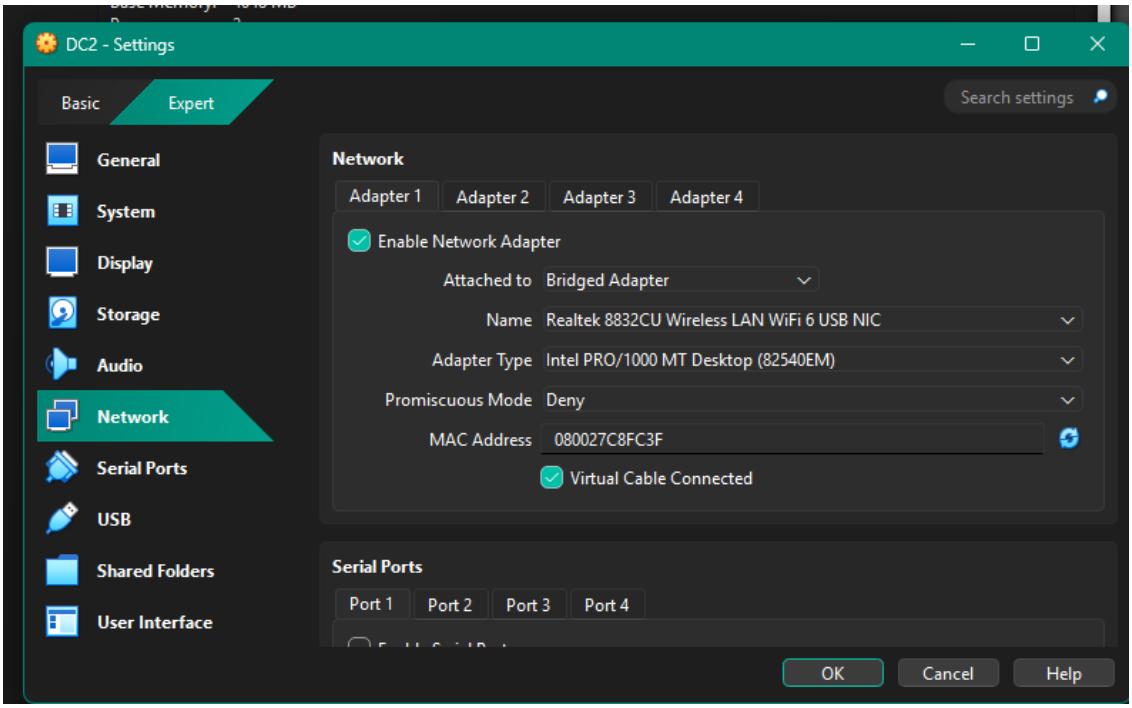


50GB hard drive space.

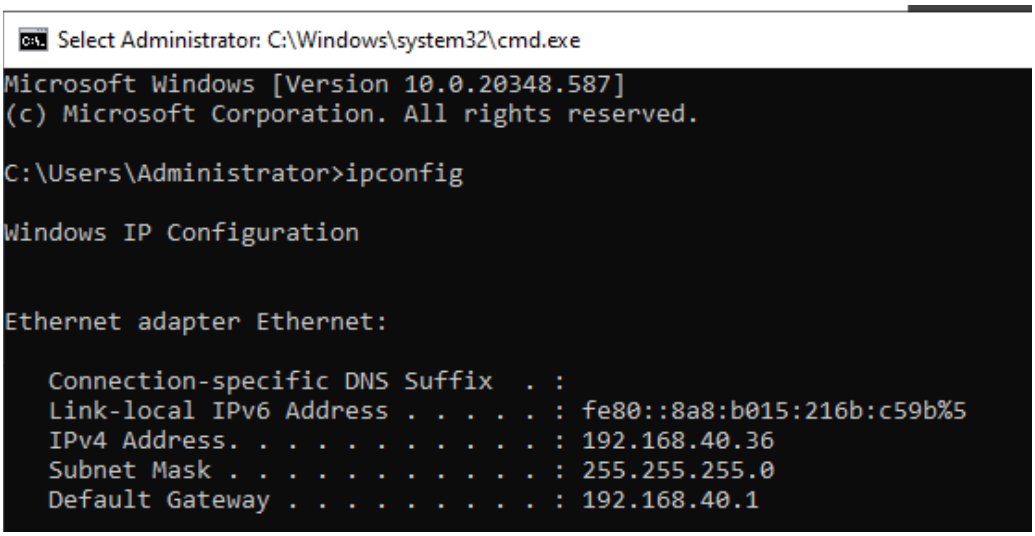


Home Lab Setup

Attach to the network adapter:



Data Center – Desktop Experience version.
Custom Installation.
Note: If VM's go black during restart phase of installation, select Machine > reset.
Setup password and login to server as administrator.
Verify IP address at the command line.

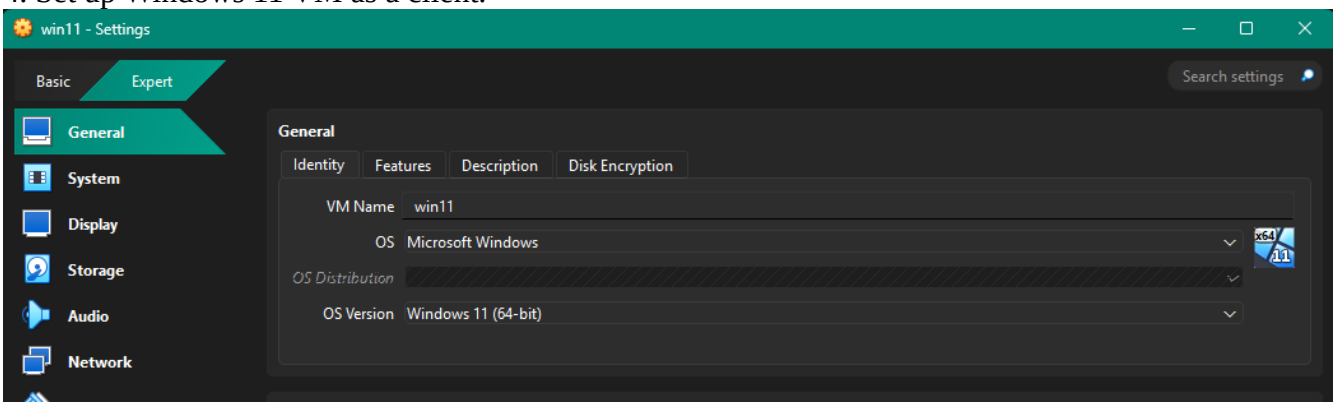


Change the system name: start > system, change server name "dclab". Restart VM.

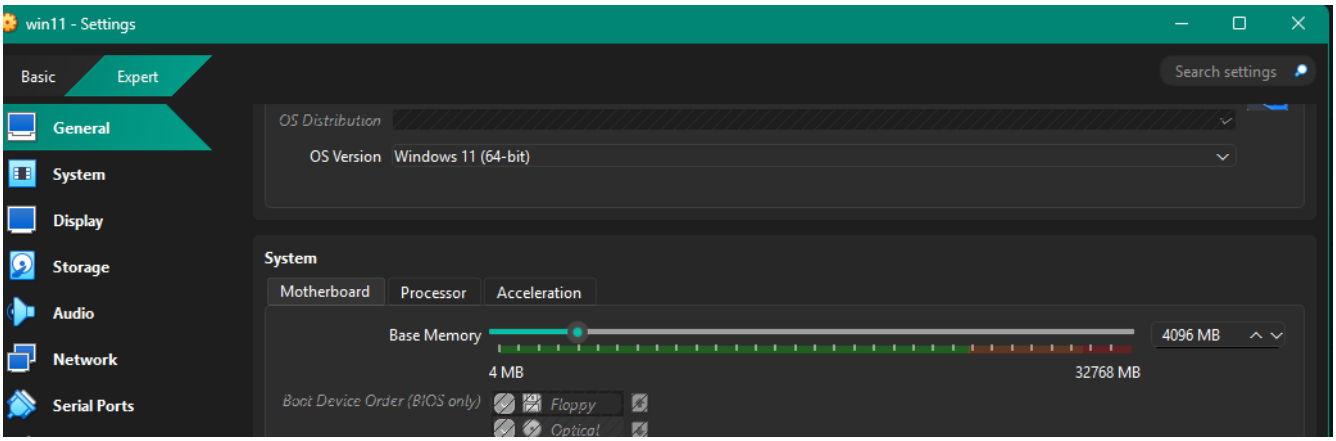
Shut down the VM to free up RAM.

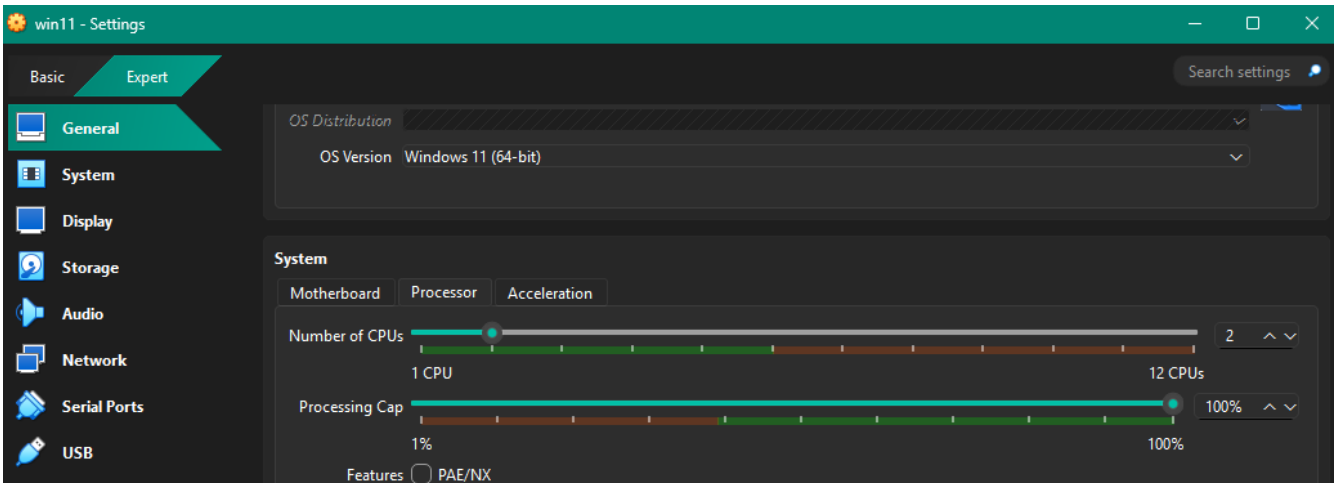
Home Lab Setup

4. Set up Windows 11 VM as a client.



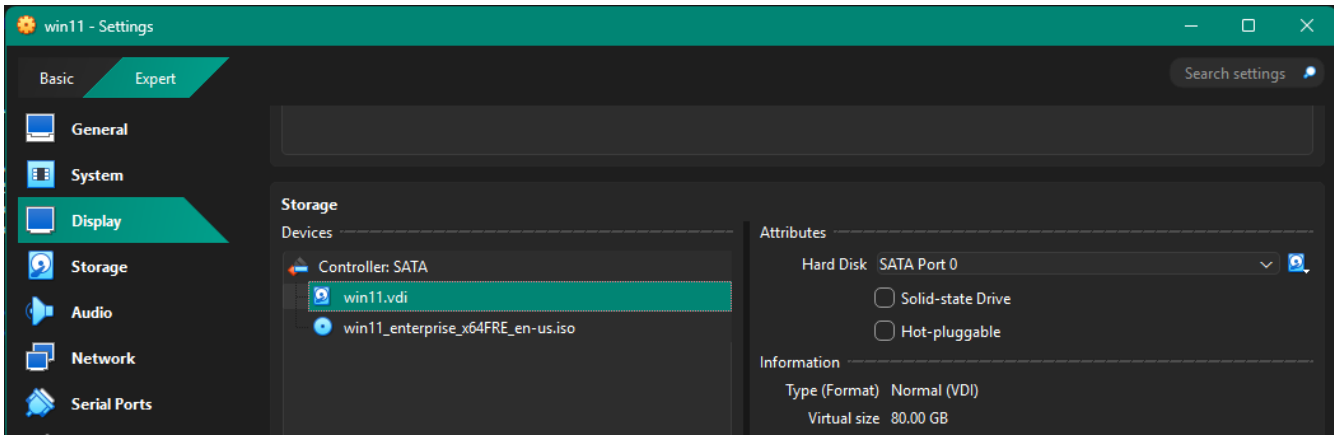
4GB RAM with 2 CPU cores



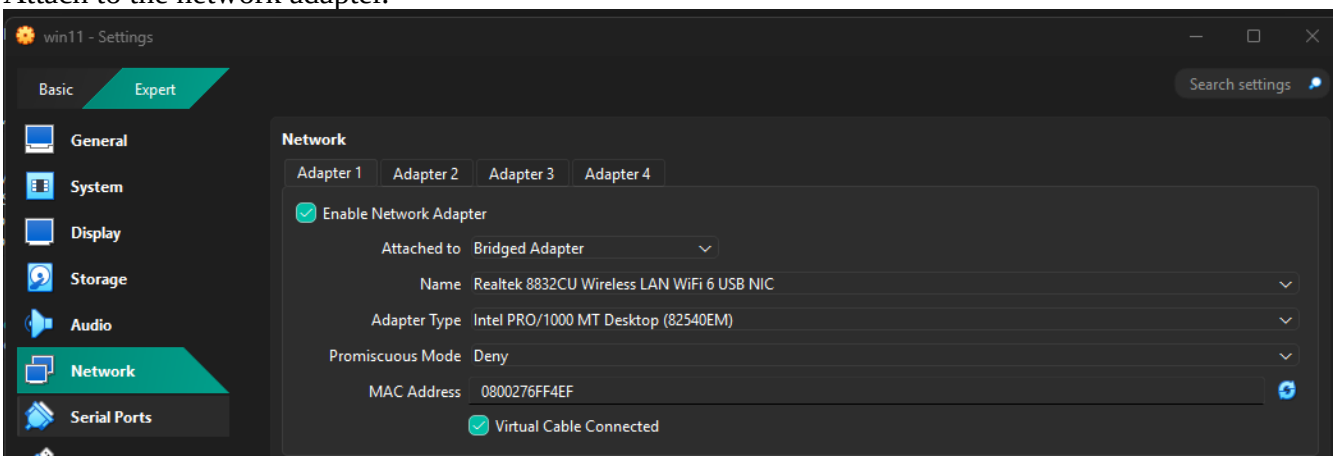


Home Lab Setup

80GB hard drive space



Attach to the network adapter.



Start with GUI.

Select Sign in options (instead of microsoft login), select the domain join.

For name, enter “admin”, enter password.

Enter security question answers.

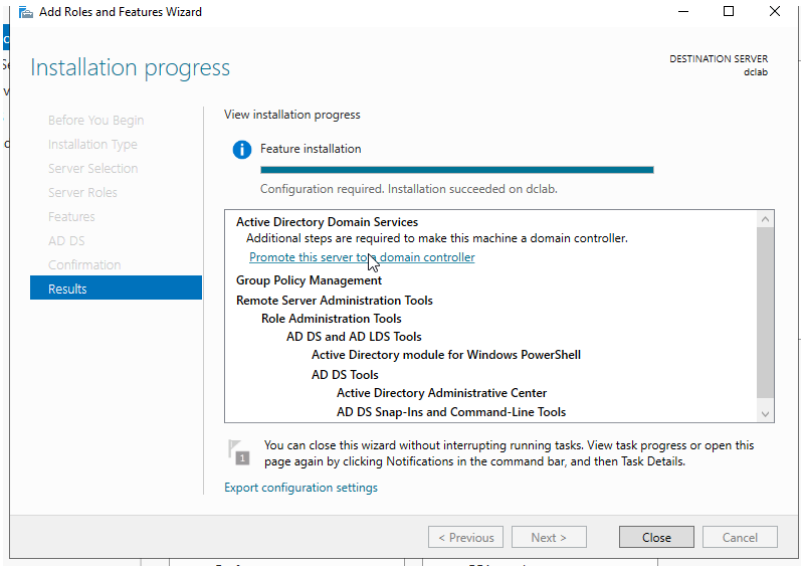
Turn off location. Complete setup.

Allow update downloads/updates to complete.

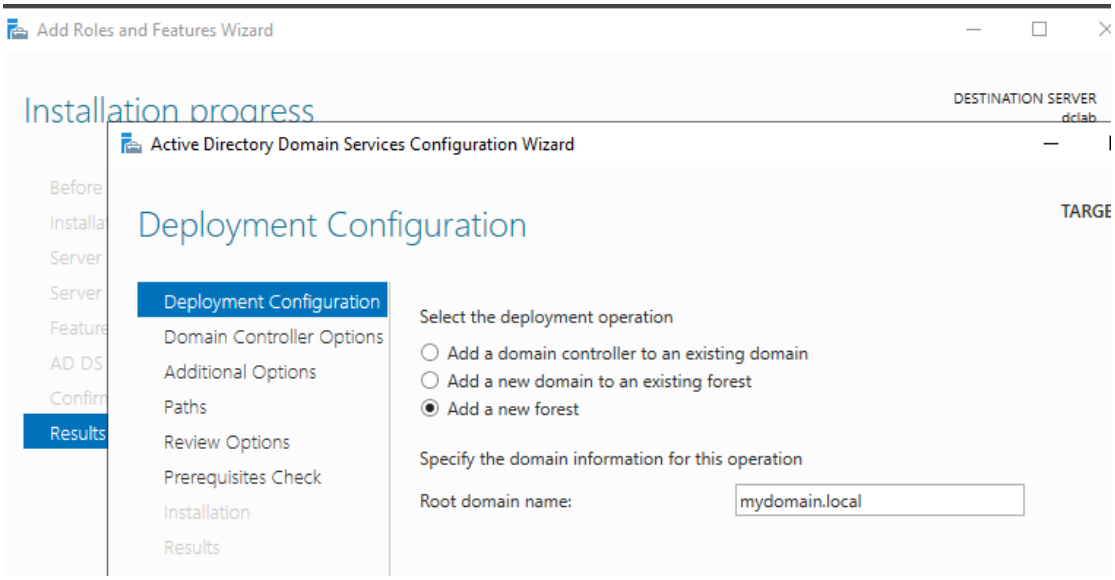
Home Lab Setup

Promote Server to Domain Controller

1. Go into the domain controller and verify the system name has been updated to dclab.
2. Go into server manager, Manage > add roles and features.
3. Under server roles, select active directory domain services.
4. Select defaults and install.
5. Once complete, click the message “promote this server to a domain controller”.



6. Add a new forest, provide a root domain name.

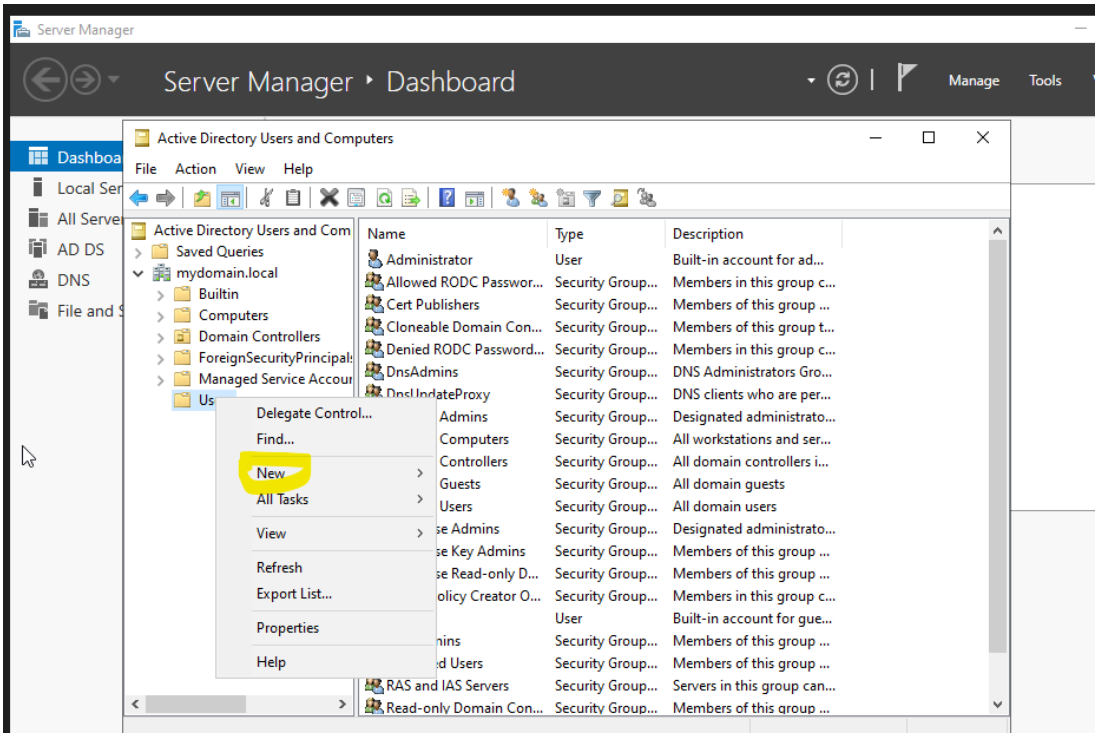


7. Accept defaults on the domain controller options and create a Directory Services Restore Mode (DSRM) password.
8. Accept defaults for the AD DS and log file locations
9. Install. Installs DNS. Restarts system.

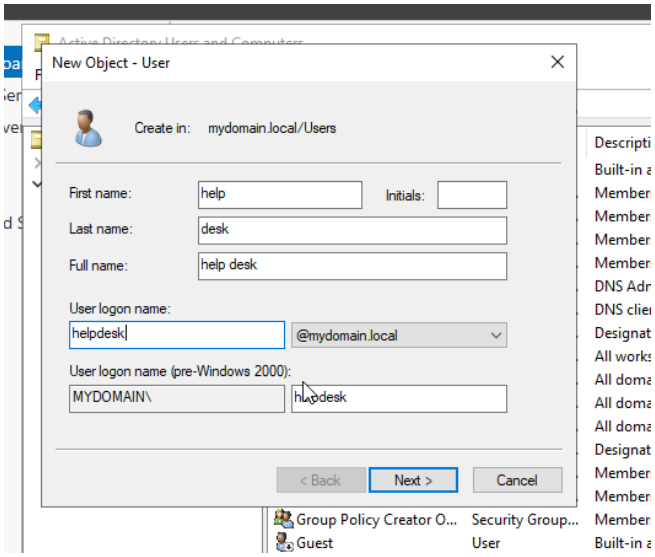
Home Lab Setup

Add a new user in Active Directory on the Domain Controller

1. Under server manager, select tools > active directory users and computers
2. Right click on users and select add new user.



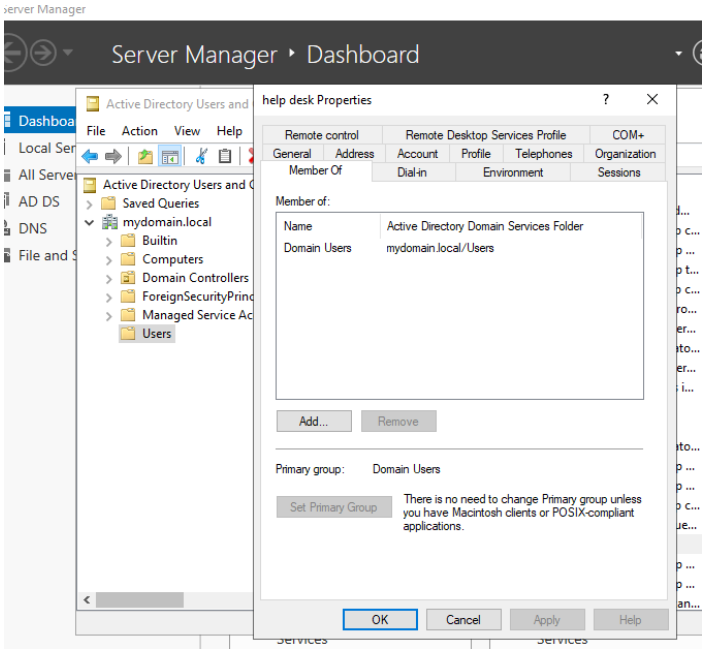
3. enter users first and last name and logon name.



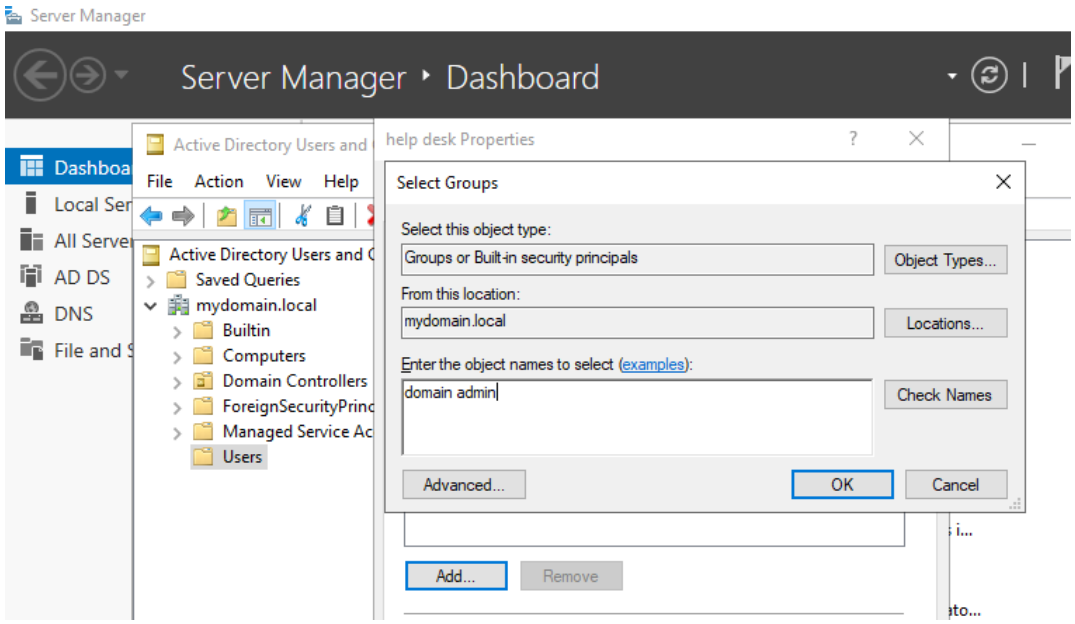
4. Select next and enter the password. Deselect the option that forces the user to change passwords before logging in.

Home Lab Setup

5. Right click on the user name, select properties, and go to the “member of” tab.



6. Select “add” and enter “domain admin”. Hit check names to find the group in the dialog.



7. Hit ok and then select apply on the next dialog box.

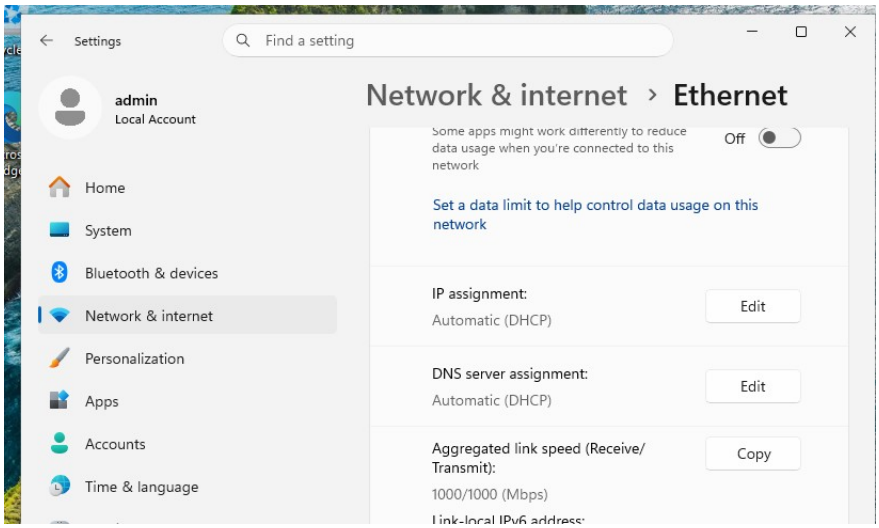
Home Lab Setup

Convert the Windows 11 VM to a Domain Managed Client

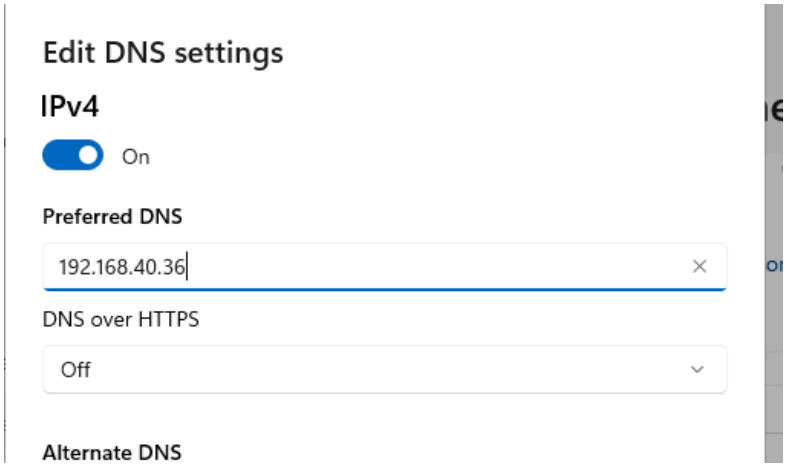
1. Change the name of the Win11 VM to win11 and restart.

2.Open the Windows 11 VM (as a workgroup) and search for ethernet settings.

3. edit the DNS Server Assignment



4. Select manual, IPV4, and enter the IP address of the DC under “preferred DNS” and save.



5. System > domain or workgroup. Select change, select member of Domain and enter “mydomain.local”. Then, when prompted, enter the helpdesk username and password in the dialog box that pops up. Once connected to the domain, the VM will restart.